

임베디드 제어(HW) 사전학습 — 10시간 계획

(Arduino / Embedded Control-HW pre-study, mapped to internship Table 1)

This plan is built directly from [표 1] 인턴십 교과목 매칭표 — specifically the 임베디드 제어(HW) row, since that’s the one that’s actually “Arduino”:

Internship table field	Value
영역 (Area)	임베디드 제어 (HW)
수업시수 (Hours)	10시간
현장 업무 내용 (On-site work)	아두이노 회로 구성, 센서 연동, 시리얼 통신
매칭 교과목 (Matching subject)	마이크로프로세서·임베디드시스템 (비교과)
학년 (Grade)	2학년
교과에서 다루는 관련 내용 (Covered content)	MCU 구조, GPIO, ADC/PWM, 인터럽트, 통신

Everything below covers exactly those five content items, plus the three on-site tasks, in **10 hours total**, split into 5 sessions of 2 hours each.

Note: the 임베디드 제어(펌웨어) row (C프로그래밍 — 변수·제어문·함수·배열·포인터) is a *separate* 10-hour block and a prerequisite for writing clean C on the Arduino. If you haven’t covered pointers/arrays/functions yet, do that first — this plan assumes basic C syntax and focuses only on the embedded/hardware-specific content.

Coverage map (curriculum item → session)

교과 내용	Covered in
MCU 구조	Session 1
아두이노 회로 구성 (on-site task)	Session 1, 2
GPIO	Session 2
ADC/PWM	Session 3
센서 연동 (on-site task)	Session 3
인터럽트	Session 4
통신	Session 5
시리얼 통신 (on-site task)	Session 5

Folder structure

```
arduino-internship-prep/
├── README.md                                <- this file
├── session-01-mcu-circuit-basics/          <- MCU 구조 + 회로 구성 (2h)
│   ├── notes.md
│   └── practice/basic_io_test.ino
├── session-02-gpio-digital-io/             <- GPIO (2h)
│   ├── notes.md
│   └── practice/button_led_gpio.ino
├── session-03-adc-pwm-sensor/              <- ADC/PWM + 센서 연동 (2h)
│   ├── notes.md
│   └── practice/sensor_pwm_control.ino
└── session-04-interrupts/                 <- 인터럽트 (2h)
```

```
| |─ notes.md
| |─ practice/interrupt_button_counter.ino
|─ session-05-serial-comm-capstone/      <- 통신 + 시리얼 통신 + integration (2h)
| |─ notes.md
| |─ schematic/circuit-diagram.svg
| |─ src/integrated_sensor_system.ino
|─ progress-tracker.md
```

How to use

1. Work session by session — each is scoped to fit in ~2 hours (theory + hands-on).
2. Build each `practice/*.ino` in Tinkercad first, then on your physical Arduino kit.
3. Session 5 is a **mini capstone**: it combines GPIO + ADC/PWM + interrupt + serial communication into one system — closely mirroring the actual 현장 업무 (아두이노 회로 구성 / 센서 연동 / 시리얼 통신) you'll be doing in the internship.
4. Tick off `progress-tracker.md` as you complete each session.

What “10 hours” means here

5 sessions × 2 hours = 10 hours, matching the table's 수업시수 exactly. If you have less time per sitting, split each session into two 1-hour blocks (notes+concept, then practice) — content stays the same either way.